פרשת וישלח

Yaakov Sent Animals To Eisav

by Bumy Goldson and Aaron Kronenfeld

These are the number of animals that יעקב sent to עשו (בראשית ל"ב:ט"ו - ט"ז):

Sheep and Goats : 200 females with 20 males (10:1 ratio)

Cows : 40 females with 10 males (4:1 ratio)

Donkeys : 20 females with 10 males (2:1 ratio)

Camels : 30 females with 30 males (1:1 ratio)

רש"י brings a מדרש that learns out from these פסוקים the amount of עונה ("time", how often they need to spend time at home with their wives) that is required from different types of workers:

Loafers (טילים) : 1 per day

Day-workers (פועלים) : 2 per week

Donkey Drivers (חמרים) : 1 per week

Camel Drivers (גמלים) : 1 per month

Sailors (ספנים) : 1 per 6 months

Sheep and Goats (who don't work) correspond to Loafers.

Cows = Day-workers

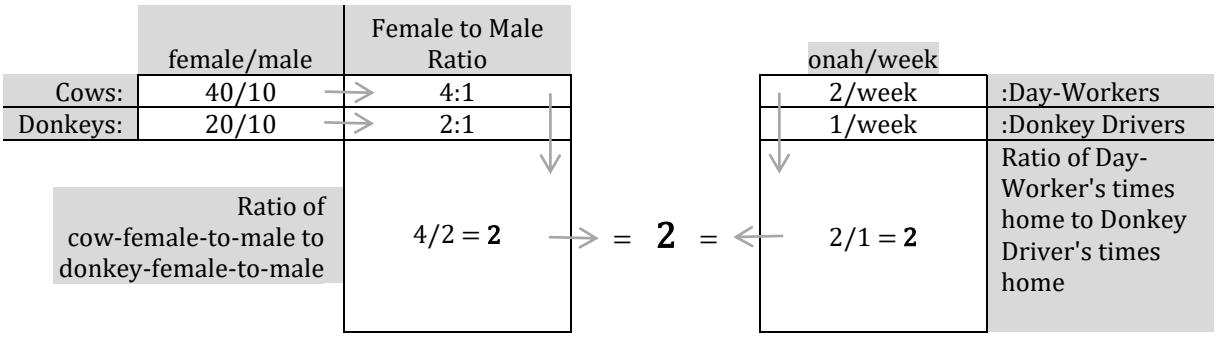
Donkeys = Donkey Drivers

Camels = Camel Drivers

Sailors don't seem to have a corresponding animal here.

רש"י says that he doesn't know how to follow this מדרש exactly. It seems the problem is that the various ratios don't match up.

Let's start with 2 that do seem to match up: Cows and Donkeys. The ratio of females these animals have is the same as the ratio of their corresponding human-workers.



So that's ok.

But the other groups don't match up. For example, between Sheep and Cows is a ratio of 10:4 $\left(= 2\frac{1}{2}\right)$ but their corresponding humans (Loafers and Day-Workers) have a ratio of 7:2 $\left(= 3\frac{1}{2}\right)$.

Let's graph it out (The 2 axis's have been adjusted so that the largest amounts between the 2 groups match up).



Doesn't match up perfectly. So, how can we learn this from that?

רש"י concludes that he thinks the פשט is not that we're learning out the specific numbers from the animals, but rather the general principle that the more one works the less he is expected to spend time at home.

Aaron Kronenfeld suggested the following תרץ.

The גמרא in בכורות (8a) gives the various gestation periods of various animals. Let's see if that is relevant to our רש"י here.

רש"י mentioned that once a female animal is pregnant she will no longer be with a male. If so then we must take into account the gestation period of each animal.

Why?

Well, let's look at sheep and cows as an example. Sheep have a gestation period of 5 months while cows have a period of 9 months. Assuming the females become pregnant more or less as soon as possible, it comes out that the male sheep is with his 10 wives every 5 months (at most, since once each one becomes pregnant they will no longer be with the male until after the baby is born, 5 months later).

On the other hand, the cows are with their 4 wives over a period of 9 months.

It comes out that you can't compare the rates between the animals unless you take the gestation period into account.

If you want to compare the sheep to the cows you need to adjust their female ratios by their gestation period.

To continue with our example. Let's put it this way. How many females are male sheep with over 9 months? Once we know that then we can compare sheep with cows and get a ratio. (In math terms, this would be similar to finding a common denominator.)

If sheep are with 10 females in 5 months, that means they are with 18 females in 9 months.

$$10 \cdot \frac{9}{5} = 18$$

That means, if a sheep were a cow (as far as gestation period is concerned) we'd expect him to have 18 wives. Instead, we find that the cow has only 4. That's a drop of $4\frac{1}{2}$ times $(18 / 4)$.

The מדרש says that due to his hard work he has less wives.

So now, let's go over to people-workers.

If we take the loafers as our starting point and assume that due to their inactivity their עונה is once a day then to find the amount due from the day-workers we need to divide "once a day" by $4\frac{1}{2}$ (which is the ratio difference we extracted from between the sheep and the cows). That comes out to a number that rounds off to zero $\left(1 \div 4\frac{1}{2} = \frac{2}{9}\right)$.

Since it doesn't make sense to assign them zero times at home we need to move on to the next unit of time after "a day", and that would be "a week". To turn that last number $\left(\frac{2}{9}\right)$ into times per week we need to multiply it by 7 (for the 7 days of the week, since we're moving from day to week) and we get $1\frac{5}{9}$ $\left(\frac{2}{9} \cdot 7\right)$, which rounds off to 2 times per week, which in fact is the amount given by חז"ל.

That's the method.

Now let's plug this formula in to the rest of the animals/workers.

$$\text{Sheep_Wives} \cdot \frac{\text{Target_Animal_Gestation_Period}}{\text{Sheep_Gestation_Period}} \div \text{Target_Animal_Wives}$$

Donkeys have 2 females, and a gestation period of 12 months. Like before, taking sheep as our starting point with 10 females in 5 months, how many would that be in 12 months?

$$\frac{10}{5} \cdot 12 = \mathbf{24}$$

All else being equal, we'd expect the donkey to have 24 females but instead we find that he only has 2. That's 12 times less.

$$24 \div 2 = \mathbf{12}$$

So now, going to people, if the loafers are home once per day then the donkey drivers should have 12 times less than that, which comes out to $\frac{1}{12}$ per day, which rounds down to zero. So like we did with the cows, let's move to the next unit of time, a week.

$$\frac{1}{12} \cdot 7 = \frac{7}{\mathbf{12}} \textbf{ per week}$$

which rounds up to **once per week**, just like the מדרש says.

Now the camel.

If a sheep has 10 females in 5 months, he should have 24 in 12 months (like we calculated with the donkey). Instead, we find the camel has only one wife, which is 24 times less than what we'd expect (if he didn't do work at all, like a sheep).

So now, carrying this over to people/workers, taking the loafer as our starting point, once a day divided by $24 = \frac{\mathbf{1}}{\mathbf{24}}$, which rounds down to zero. Moving to the next unit of time, a week $\left(\frac{1}{24} \cdot 7\right)$, gives us $\frac{\mathbf{7}}{\mathbf{24}}$, which also rounds down to zero. Moving on to the next unit of time, a month, (taking a month as 4 weeks) it comes out to $1\frac{1}{6}$ which rounds down to **once per month**, which is exactly what the מדרש says! (The חשבון works out even if you use a more accurate figure for how many weeks are in a month.)

T	**F**	**G**	**E**	**R**	**D**		**W**		**M**		**O**	
Target Animal	**F**emales	**G**estation Period (in months)	**E**xpected Females, based on Sheep (G ÷ Sheep Gestation * Sheep_ Females)	**R**atio of Expected to Actual (E ÷ F)	Onah per **D**ay (1 / **R**)	Rounds to... (per day)	Move to next Unit, week (**D** * 7)	Rounds to... (per week)	Move to next Unit, month (**W** * 4)	Rounds to... (per month)	Onah	Corresponding Worker
Sheep	10	5	5	1	1	1	-	-	-	-	1/day	**Loafers**
Cows	4	9	18	$4\frac{1}{2}$	$\frac{2}{9}$	0 →	$1\frac{5}{9}$	2	-	-	2/week	**Day-Workers**
Donkeys	2	12	24	12	$\frac{1}{12}$	0 →	$\frac{7}{12}$	1	-	-	1/week	**Donkey Drivers**
Camel	1	12	24	24	$\frac{1}{24}$	0 →	$\frac{7}{24}$	0 →	$1\frac{1}{6}$	1	1/month	**Camel Drivers**

Now, we still don't have a source for sailors.

I would like to suggest that perhaps the sailors correspond to the snake, which the גמרא in בכורות gives 7 years as the gestation period.

Why the snake?

Well, our פסוק here in וישלח doesn't give us a corresponding animal from which to learn. So, perhaps חז"ל took the animal with the greatest gestation period in order to learn what is at least the minimal amount one could be חייב (המוציא מחברו עליו הראיה). Or maybe we can pin it on the פסוק in משלי (30:19) דרך נחש עלי צור דרך אניה בלב ים *"The way of the snake upon the rock, the way of a ship in the heart of the sea"*.

(And even though the גמרא there says that snakes <u>will</u> be with their wives even when they're pregnant, and even though we don't have a פסוק telling us how many wives a snake has, we could say that for this case it is a לימוד בעלמא.)

Applying our formula:

T	**F**	**G**	**E**	**R**	**D**		**W**		**M**		**Y**		**O**	
Target Animal	**F**emales	**G**estation Period	**E**xpected Females, based on Sheep (G ÷ Sheep Gestation * Sheep_ Females)	**R**atio of Expected to Actual (E ÷ F)	Onah per **D**ay (1 / **R**)	Rounds to... (per day)	Move to next Unit, week (**D** * 7)	Rounds to... (per week)	Move to next Unit, month (**W** * 4)	Rounds to... (per month)	Move to next Unit, year (**M** * 12)	Rounds to... (per year)	Onah	Corresponding Worker
Sheep	10	5 months	5	1	1	1	-	-	-	-	-	-	1/day	**Loafers**
Snake	1	7 years (12 * 7 = 84 months)	168	168	$\frac{1}{168}$	0 →	$\frac{1}{24}$	0 →	$\frac{1}{6}$	0 →	$\frac{12}{6}$	2	2/year (1 per 6 months)	**Sailors**

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